Iron depletion and anaemia: prevalence, consequences, diagnostic and therapeutic implications in a developing Pakistani population

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Summary

The prevalence of iron deficiency anaemia in Karachi, Pakistan, was examined in a group of 100 children, aged 2 to 6 years, and 200 adult females, aged 17 to 35 years. Anaemia, defined as a haemoglobin level less than 11.5 g/dl, was present in 47% of the children and 30% of the adult females. The majority of subjects had normal red cell counts and MCV, but haematocrit and MCH were below normal in one-quarter of the children and one-third of the adult females. More than half of both children and adult females had serum ferritin levels below normal. Many also had elevated iron binding capacity and reduced serum iron levels. Iron deficiency may be superimposed on folate deficiency in this population. The potentially serious consequences and higher prevalence of iron deficiency anaemia in children and women of childbearing age demand greater efforts toward diagnosis and eradication in developing countries.

Key words: Anaemia, hypochromic — health surveys

Introduction

Iron deficiency anaemia can have adverse effects on both mental and physical performance.[1,2,6,9] These consequences have more serious implications in developing countries than in the developed nations. Anaemia is more prevalent in developing nations, although its occurrence in Western countries is still significant.[1] Anaemia is particularly common in children and women of child-bearing age.[8]

This paper presents a discussion of the prevalence of iron deficiency anaemia in a local population in Karachi, Pakistan. Anaemia was assessed using serum ferritin, which has proven to be a useful indicator of iron deficiency states.[3,11]

Subjects and methods

The study population was drawn from Karachi, the largest city in Pakistan and one which contains persons from all provinces of Pakistan. A total of 300 subjects was included in the study: 100 children (49 males and 51 females), aged 2 to 6 years; 100 females, aged 17 to 21 years, with no history of pregnancy; and 100 females, aged 25 to 35 years, with at least 1 prior pregnancy. Most individuals were from low-income or middle-income groups; only 10% were from the higher-income group.

Blood samples were collected for measurements of serum ferritin, serum iron, total iron binding capacity (TIBC), red cell indices and films. Serum ferritin was

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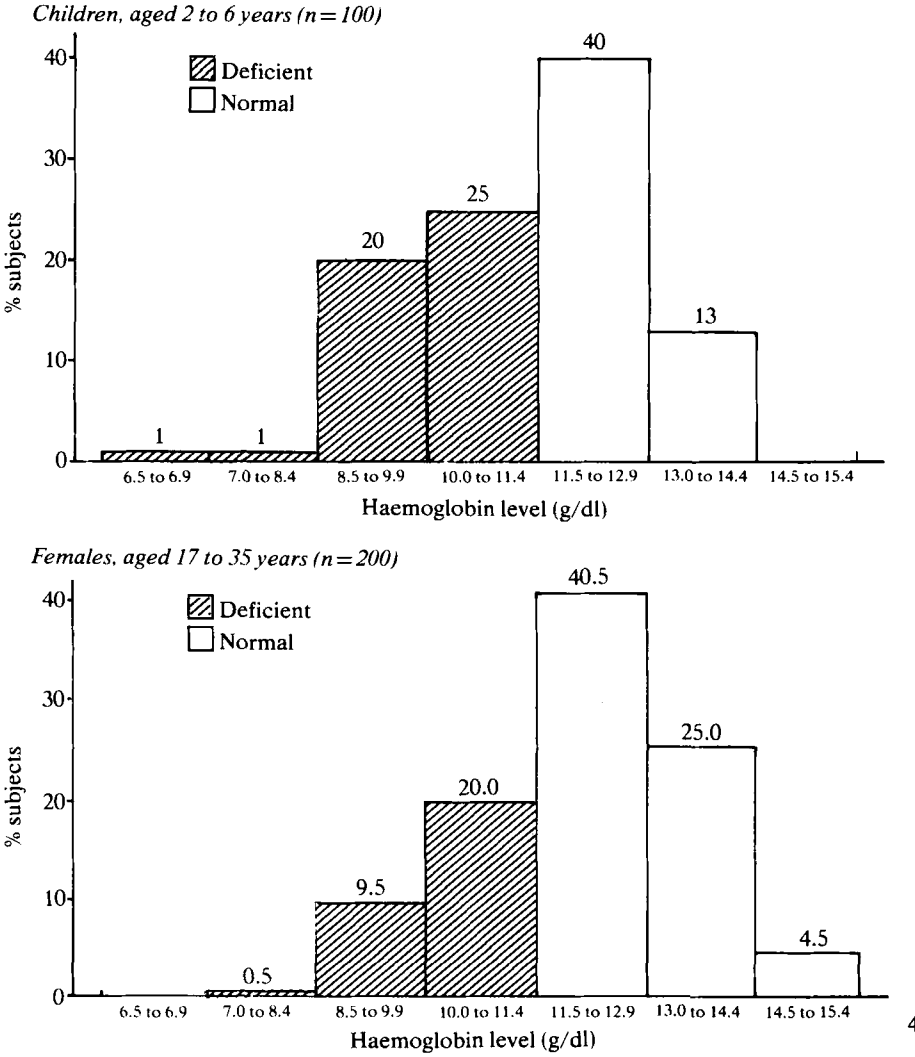
measured with a 'Ferrizyme' kit (Abbott Laboratories) which utilizes an enzyme immunoassay. Measurements of serum iron and TIBC were accomplished with Roche kits. Absolute red cell indices were done on an ERMA PC-604 Blood Cell Counter. Films were made manually and stained by Leishman's stain.

Results for haemoglobin, mean corpuscular haemoglobin (MCH), mean corpuscular haemoglobin concentration (MCHC), mean corpuscular volume (MCV), red cell count, serum ferritin, serum iron and TIBC were analyzed by subject group. All statistical analyses were done using Statistical Analyses System (SAS) software.[10]

Results

The distributions of haemoglobin level values for children and adult females are shown in Figure 1.

Figure 1. Distribution of haemoglobin level values for children and adult females in the population studied: percentage of subjects



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Forty-seven percent of the children and 30% of the adult females were anaemic (haemoglobin below 11.5 g/dl). However, only 2% of the children and 0.5% of adult females had haemoglobin levels below 8.5 g/dl. Distributions for the remaining haematological variables are summarized in Table I.

Table I. Distribution of haematological variables in the population studied: percentage of subjects

Variable	Children, 2 to 6 years (n = 100)		Females, 17 to 35 years (n = 200)	
	% below normal	% above normal	% below normal	% above normal
Red cell count	7	93	8	92
Haematocrit	28	72	31	69
MCV	9	91	14.5	85.5
MCH	25	75	34	66
MCHC	28	72	18	82

Relatively small proportions of children and adult females had below-normal values for red cell count and MCV. However, for haematocrit and MCH, approximately one-quarter of the children and one-third of the adult females had values below the normal range. In addition, the mean value of MCH for adult females was below the lower limit of normal.

Table II shows the mean values for iron states in the two groups of subjects. In both groups, the geometric mean for serum ferritin was below the lower limit of normal (10 ng/ml). In addition, mean serum iron for children was slightly below the lower limit of normal (37 μg/ml).

Table II. Iron states in the two groups of subjects: mean (±S.D.) values

Variable	Children, 2 to 6 years (n = 100)	Females, 17 to 35 years (n = 200)
Serum ferritin (ng/ml):		
Arithmetic mean	21.67±1.89	21.48±25.84
Geometric mean	6.17	9.39
Serum iron (μg/ml)	36.56±23.49	44.78±20.21
TIBC (μg/dl)	377.48±127.49	331.73±133.66

The distributions of serum ferritin values for children and adult females are depicted in Figure 2.

Slightly more than half of both children and adult females had below-normal values, and roughly one-quarter of each group had serum ferritin values in the lower range of normal.

Table III summarizes the prevalence figures for anaemia and iron deficiency in the study population. Iron binding capacity was elevated in 44% of the children and 32% of adult females, and serum iron was below 37 μg/ml in 58% of children and 42% of adult females.

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Figure 2. Distribution of serum ferritin level values for children and adult females in the population studied: percentage of subjects

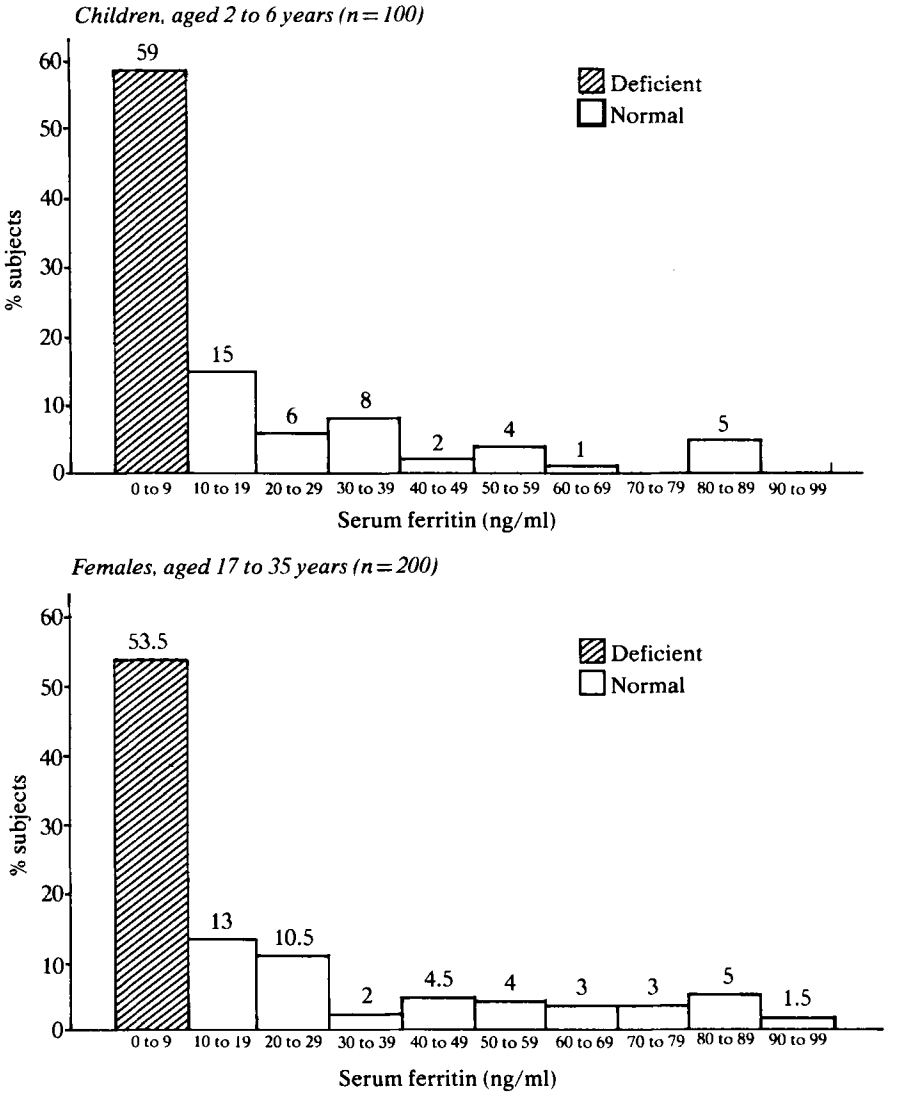

Table III. Prevalence of anaemia and iron deficiency in the population studied: percentage of subjects

Variable	Children, 2 to 6 years (n=100)	Females, 17 to 35 years (n=200)
Haemoglobin <11.5 g/dl	47	30
Serum ferritin <10 ng/ml	59	54
Serum iron <37 μg/ml	58	42
TIBC >400 μg/dl	44	32

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Discussion

The results of this study show that 36% of the total study population were clearly anaemic. Of the remaining subjects who were not anaemic, 45% had evidence of iron depletion, as indicated by serum ferritin values. A substantial proportion of subjects showed elevated iron binding capacity and reduced serum iron values, as well. Forty-seven percent of the children were anaemic, compared to 30% of the adult females. In contrast, both groups had roughly similar proportions of individuals with low serum ferritin (58% of children and 54% of adult females). In the children, iron deficiency may be superimposed on folate deficiency; a greater proportion of children had increased MCV versus MCH. The local diet consists primarily of bread, rice and vegetables. While some of the vegetables, such as spinach, are rich in iron, it is not in an absorbable form. In addition, bread and rice contain high levels of phytates and other substances which prevent iron absorption.[9] The normal mean MCV for children is 81 fl; 75% of children in this study had values above this level. Of those children with MCV values greater than the normal mean, 35% had haemoglobin values below 11.5 g/dl. Further studies of folate deficiency in this population are necessary.

Biochemical markers delineate three stages of iron deficiency. In the first stage, haemoglobin, serum iron and iron binding capacity are normal, but serum ferritin is decreased. Serum ferritin and serum iron are low in the second stage, while iron binding capacity is elevated and haemoglobin is normal. In the third stage of iron deficiency with anaemia, iron binding capacity is elevated and serum iron, serum ferritin and haemoglobin are decreased. In most developing health-care setups, including ours in Pakistan, iron deficiency is routinely diagnosed based on a low haemoglobin, hypochromia and microcytosis. Practically all laboratories report these end-stage changes, but these values do not help in the prevention or early diagnosis of iron depletion. As this study demonstrates, low levels of serum ferritin correlate well with iron depletion, particularly given the background of the local poor dietary iron intake and the socioeconomic situation.

The consequences of iron deficiency anaemia include poor mental performance in infants[1] and school children,[6,9] and poor physical performance in workers. Concentration and performance are better in non-anaemic individuals than in those persons with iron deficiency. Iron requirements are greater in pregnancy, even in the early gestational period.[4] Animal studies have shown metabolic and functional deficits associated with anaemia.

Recognition of the seriousness of the effects of anaemia has led to efforts at eradication of iron deficiency in the United States and Europe. A study by Addy[1] in the United Kingdom demonstrated low serum ferritin levels in 23% of white children and 45% of Asian children. The data presented here reveal an even more extensive problem in Pakistan. The mistaken concept, prevalent in Pakistan, that haemoglobin levels of 10 to 11 g/dl are perhaps normal for that population should be discouraged, since clinical observations reveal complications due to low haemoglobin levels occur very frequently. Subjects in our study who had such low levels of haemoglobin were largely iron depleted, as demonstrated by low serum ferritin levels in 91% of girls aged 17 to 21 years, 57% of married females aged 25 to 35 years,

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89% of male children and 68% of female children. Much effort will be required to prevent the consequences of iron deficiency anaemia[5] in the Pakistani population and that of other developing nations.

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